Chapter 5 • Linear Equations and Inequalities

Solved Review Exercise 5

MCQs

1. Which is a linear equation?
a) $5x>7$
b) $4x-2<1$
c) $2x+1=1$
d) $4=1+3$
Ans: c
It is an equation, and the variable has power 1.

2. The solution of $5x-10=10$ is:
a) 0
b) 50
c) 4
d) -4
Ans: c
$5x=20$, so $x=4$.

3. If $7x+4<6x+6$, then x belongs to:
a) $(2,\infty)$
b) $[2,\infty)$
c) $(-\infty,2)$
d) $(-\infty,2]$
Ans: c
Subtract $6x+4$ to get $x<2$.

4. A vertical line divides the plane into:
a) a left half-plane only
b) a right half-plane only
c) a full plane
d) two half-planes
Ans: d
The two sides are left and right half-planes.

5. The equation formed from a linear inequality is called:
a) a cubic equation
b) an associated equation
c) a quadratic equation
d) a feasible region
Ans: b
Replace the inequality sign by equality to get the boundary.

6. $3x+4<0$ is:
a) an equation
b) an inequality
c) not an inequality
d) an identity
Ans: b
It contains an inequality sign.

7. A corner point is also called:
a) a code
b) a vertex
c) a curve
d) a region
Ans: b
Corner point and vertex are synonyms here.

8. (0,0) is a solution of which inequality?
a) $4x+5y>8$
b) $3x+y>6$
c) $-2x+3y\leq 0$ (corrected sign)
d) $x+y>4$
Ans: c
With ≤, option (c) gives $0\leq 0$. The printed strict < would make no option correct.

9. The solution region restricted to the first quadrant is called:
a) objective region
b) feasible region
c) solution function

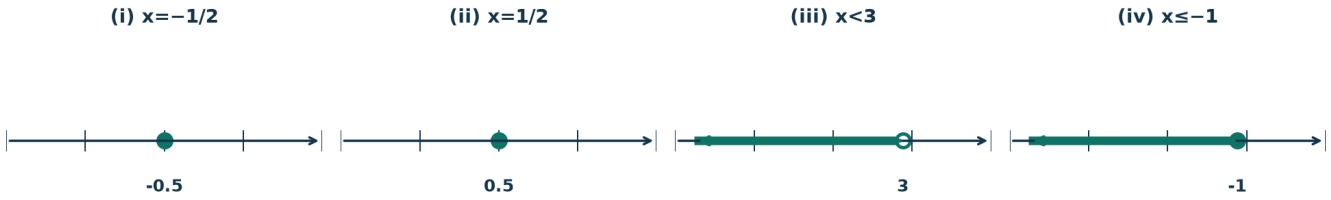
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d) constraints region
Ans: b
This is the feasible region.

10. A function to be maximized or minimized is called:
a) a solution function
b) an objective function
c) a feasible function
d) none of these
Ans: b
It is the objective function.

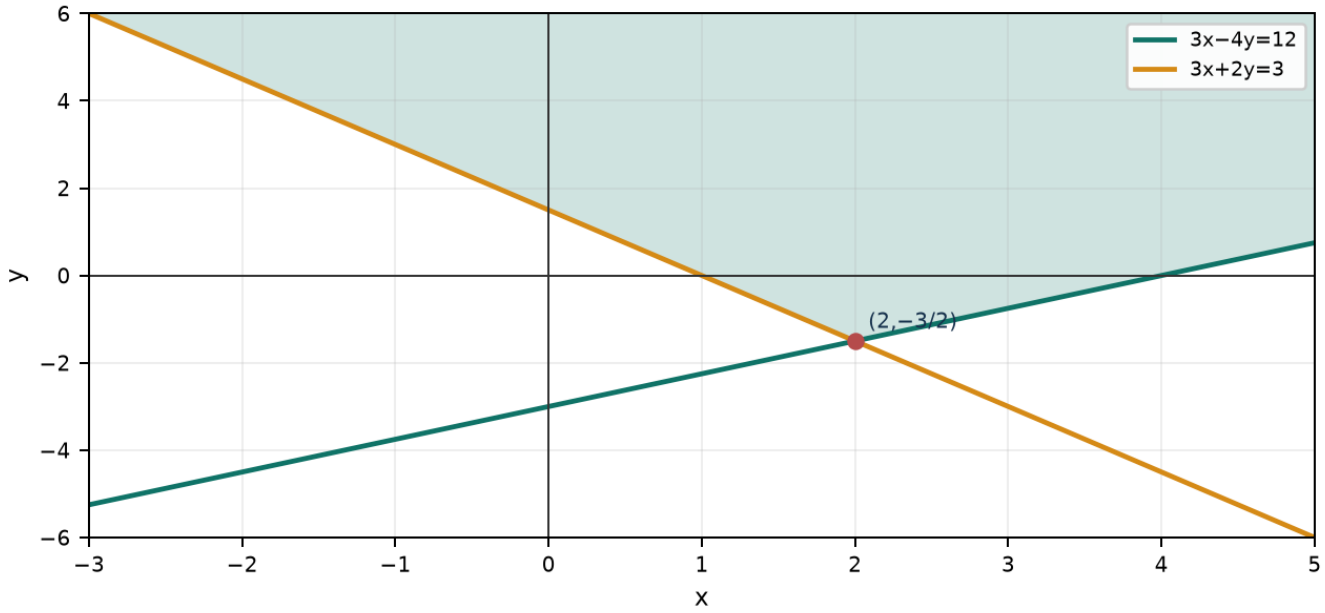
2. Solve and represent on a real line.

(i) $x+5=3-3x$
$\Rightarrow 4x=-2$
$\Rightarrow x=\frac{-1}{2}$
(ii) $4x+5=8-2x$
$\Rightarrow 6x=3$
$\Rightarrow x=\frac{1}{2}$
(iii) $3x+7<16$
$\Rightarrow 3x<9$
$\Rightarrow x<3.$
$(-\infty,3)$
(iv) $5x-15\geq 16x-4$
$\Rightarrow -11\geq 11x$
$\Rightarrow x\leq -1.$
$(-\infty,-1]$.



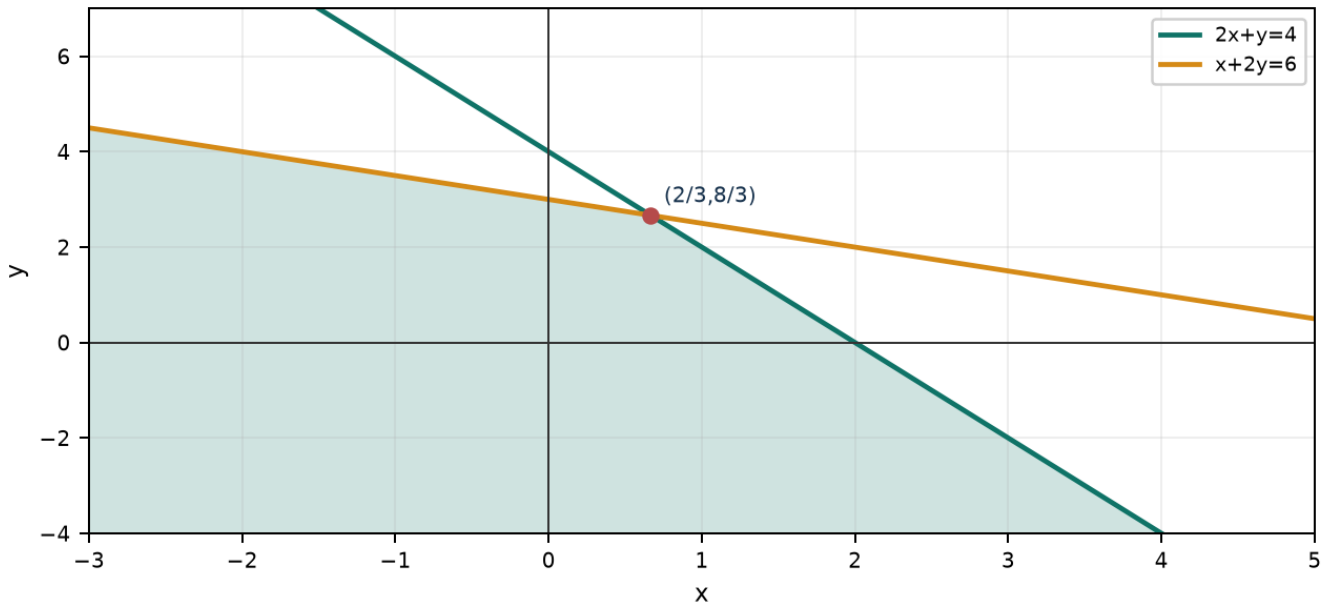
x	3x−4y=12	3x+2y=3
-2	-4.5	4.5
0	-3	1.5
2	-1.5	-1.5
4	0	-4.5

Review Exercise 5 Q3(i): simultaneous region



x	2x+y=4	x+2y=6
-2	8	4
0	4	3
0.6667	2.6667	2.6667
2	0	2

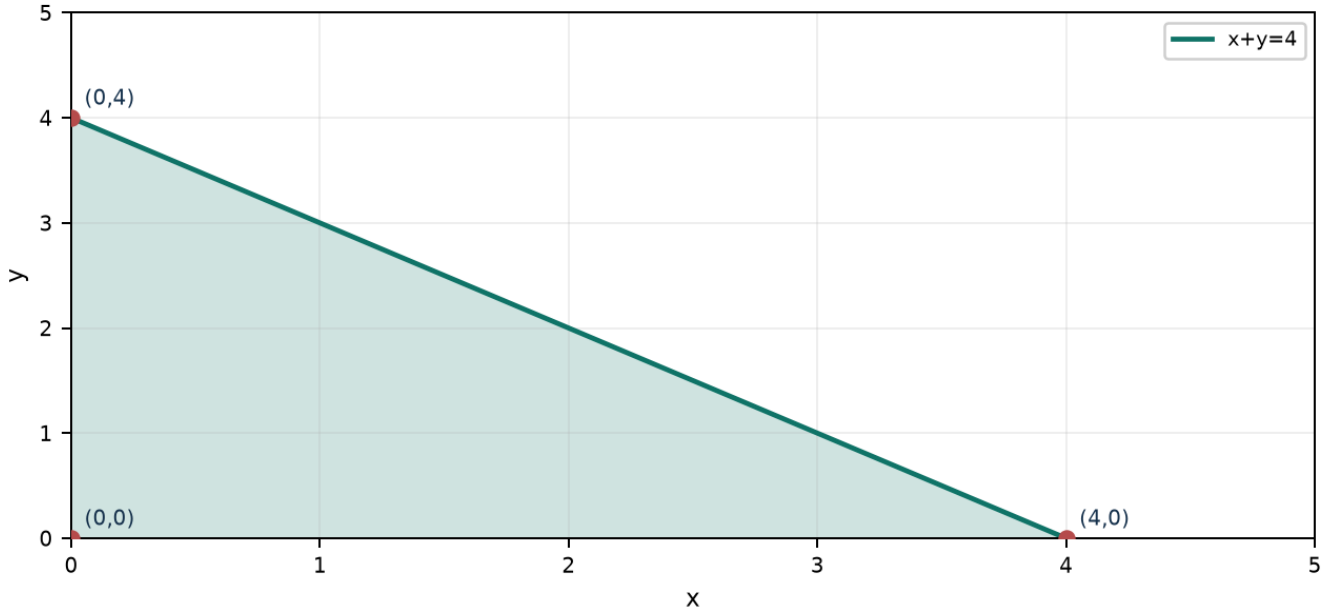
Review Exercise 5 Q3(ii): simultaneous region



feasible vertex	g=x+4y
(0,0)	0
(4,0)	4

feasible vertex	g=x+4y
(0,4)	16

Review Exercise 5 Q4: feasible triangle



feasible vertex	f=3x+5y
(3,0)	9
(3/2,1/2)	7
(0,2)	10

Review Exercise 5 Q5: lower envelope of feasible region

